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Patent Public Search | Text View

United States Patent Application Publication

20250277926

Kind Code

A1

Publication Date

September 04, 2025

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TERMINAL BLOCK AND I/O UNIT

Abstract

A terminal block includes a plurality of insertion sections each provided with a connection terminal that is electrically connected to an external terminal; a plurality of lighting sections provided corresponding to the plurality of insertion sections; a plurality of light guide members that are provided corresponding to the plurality of lighting sections and that direct light emitted by a plurality of light emitting elements to the plurality of lighting sections to cause the plurality of lighting sections to light; and at least one connecting member connecting the plurality of light guide members.

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Family ID:	88518125
Appl. No.:	18/858671
Filed (or PCT Filed):	April 28, 2022
PCT No.:	PCT/JP2022/019312

Publication Classification

Int. Cl.: F21V8/00 (20060101)

U.S. Cl.:

Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a terminal block, and an I/O unit including the terminal block.

BACKGROUND ART

[0002] JP 2001-125607 A discloses an example of an I/O unit (signal transmission device). The signal transmission device includes a display unit. The display unit includes a light emitting diode (LED) that is turned on. The display unit (lighting unit) is disposed on the front surface of the signal transmission device.

SUMMARY OF THE INVENTION

[0003] One of the parts that form the lighting unit of the I/O unit is a light guide member that guides light emitted by the light source. The light guide member is, for example, a light pipe.

[0004] The I/O unit may include a plurality of lighting units. In this case, a plurality of light guide members are provided in the I/O unit in accordance with the number of the lighting units. However, as the number of the light guide members provided in the I/O unit increases, it takes more time to assemble the light guide members to the I/O unit.

[0005] The present invention has the object of solving the aforementioned problem.

[0006] According to a first aspect of the present invention, there is provided a terminal block comprising: a plurality of insertion-receiving portions into which a plurality of external terminals are inserted, respectively, and which are provided respectively with connection terminals each electrically connected to a corresponding one of the inserted external terminals; a plurality of lighting portions provided corresponding to the plurality of insertion-receiving portions, respectively; a plurality of light guide members provided corresponding to the plurality of lighting portions, respectively, and each configured to light a corresponding one of the plurality of lighting portions by guiding, to the corresponding one of the plurality of lighting portions, light emitted by a corresponding one of a plurality of light emitting elements; and at least one coupling member coupling the plurality of light guide members.

[0007] According to a second aspect of the present invention, there is provided an I/O unit comprising: the terminal block according to the first aspect; the plurality of light emitting elements; and a communication processing circuit configured to control input/output of signals to/from the plurality of external terminals connected to the terminal block, and control light emission of the plurality of light emitting elements.

[0008] According to the present invention, an increase in the operating time for assembling the light guide members to the terminal block of the I/O unit is suppressed.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. 1 is a diagram showing a communication system according to a reference example;

[0010] FIG. 2 is a perspective view showing a front portion and a left side portion of an I/O unit according to the reference example;

[0011] FIG. 3 is a diagram for explaining the internal configuration of a connector unit according to the reference example;

[0012] FIG. 4A is a perspective view showing a front portion and a left side portion of an I/O unit

according to an embodiment, and FIG. 4B is an exploded view of the I/O unit of FIG. 4A; [0013] FIG. 5 is a diagram for explaining the internal configuration of a connector unit according to the embodiment; [0014] FIG. 6A is a diagram showing a connector unit according to a first modification, and FIG. 6B is a diagram showing another connector unit according to the first modification; and [0015] FIG. 7 is a diagram showing a connector unit according to a second modification.

DETAILED DESCRIPTION OF THE INVENTION

Embodiment

[0016] FIG. 1 is a diagram showing a communication system **100** according to a reference example.

[0017] Note that a plurality of arrows shown in each drawing indicate respective directions (front, rear, left, right, up, and down) used in the following description. An up-down direction (a first direction), a left-right direction (a second direction), and a front-rear direction are orthogonal to each other. The left-right direction and the front-rear direction are parallel to a horizontal plane, for example.

[0018] The communication system **100** is a system for transmitting signals between a controller and a device **104**. The controller is, for example, a numerical controller that controls a machine tool. The device **104** is, for example, an actuator, a detector, or the like provided in the machine tool. The controller is not shown in the drawing. FIG. 1 shows only a part including an external terminal **104a** of the device **104**.

[0019] The communication system **100** includes a plurality of I/O units (input/output units) **108**. The plurality of I/O units **108** are consecutively arranged on a predetermined installation member **110**. The number of the I/O units **108** to be consecutively arranged is not particularly limited.

[0020] The installation member **110** is, for example, a DIN rail. The extending direction of the installation member **110** illustrated in FIG. 1 is the left-right direction. That is, the installation member **110** illustrated in FIG. 1 is long in the left-right direction.

[0021] The plurality of I/O units **108** are attached to the front portion of the installation member **110**. In the illustration of FIG. 1, the arrangement direction of the plurality of I/O units **108** coincides with the extending direction of the installation member **110**.

[0022] FIG. 2 is a perspective view showing a front portion **108F** and a left side portion **108L** of the I/O unit **108** according to the reference example.

[0023] The I/O unit **108** includes a connector unit **112** and a communication processing unit **114**.

[0024] The communication processing unit **114** includes a housing **114a**, a plurality of contactors **20**, and a control module **26**. The housing **114a** accommodates the plurality of contactors **20** and the control module **26**. However, a part of each contactor **20** is exposed to the outside at the left side portion **108L** or a right side portion **108R** of the I/O unit **108**.

[0025] The plurality of contactors **20** are a terminal group electrically connected to another I/O unit **108** adjacent to the I/O unit **108**.

[0026] The plurality of contactors **20** are arranged on each of the left side portion **108L** and the right side portion **108R**. The plurality of contactors **20** on the right side portion **108R** are arranged symmetrically with the plurality of contactors **20** on the left side portion **108L**, for example. All of the contactors **20** are electrically connected to the control module **26**.

[0027] The control module **26** is a module including a communication processing circuit **22** and a plurality of light emitting elements **24**.

[0028] The communication processing circuit **22** is an electric circuit that performs processing necessary for establishing communication between the controller and the device **104**. The communication processing circuit **22** includes, for example, a control integrated circuit (IC). The communication processing circuit **22** is mounted on, for example, a resin board.

[0029] The communication processing circuit **22** is electrically connected to the plurality of contactors **20**. In this case, the communication processing circuit **22** is electrically connected to

another unit disposed on the left side of the I/O unit **108** via the plurality of contactors **20** disposed on the left side portion **108L**. Further, the communication processing circuit **22** is electrically connected to another unit disposed on the right side of the I/O unit **108** via the plurality of contactors **20** disposed on the right side portion **108R**.

[0030] The communication processing circuit **22** is connected to a plurality of the devices **104** via the connector unit **112**. Therefore, in FIG. **1**, the device **104** is electrically connected to the controller via at least one I/O unit **108**. The controller and the device **104** can input and output signals via at least one communication processing circuit **22**. The connector unit **112** will be described in more detail later.

[0031] The plurality of light emitting elements **24** include, for example, a plurality of light emitting diodes (LEDs). Each light emitting element **24** is disposed on a board, for example, in the same manner as the communication processing circuit **22**. Each light emitting element **24** can emit light under the control of the communication processing circuit **22**. The plurality of light emitting elements **24** are provided corresponding to a plurality of lighting portions **34** provided on the connector unit **112**.

[0032] The connector unit **112** is a terminal block device for connecting the external terminal **104a** (see FIG. **1**) of the device **104**. The connector unit **112** includes the front portion **108F** of the I/O unit **108**. That is, the connector unit **112** is disposed forward of the communication processing unit **114**. The connector unit **112** includes a plurality of insertion-receiving portions **28**, a plurality of release operating elements **32**, and the plurality of lighting portions **34**.

[0033] Each of the plurality of insertion-receiving portions **28** is a portion to which the external terminal **104a** is connected. Each of the insertion-receiving portions **28** includes, for example, an insertion opening into which the external terminal **104a** is inserted. The plurality of insertion-receiving portions **28** are provided in the front portion **108F**.

[0034] The plurality of release operating elements **32** are an operating element group provided corresponding to the plurality of insertion-receiving portions **28**. Each of the plurality of release operating elements **32** is, for example, a pusher. In response to an operation of an operator, each of the release operating elements **32** releases the connection between the corresponding insertion-receiving portion **28** and the external terminal **104a** connected to that insertion-receiving portion **28**.

[0035] The plurality of release operating elements **32** are disposed on the inner side of the connector unit **112** (see also FIG. **3**). However, at least a part of each release operating element **32** is exposed to the outside so that the operator can operate the release operating element **32**. For example, a part of each release operating element **32** is exposed to the outside at the front portion **108F**.

[0036] The plurality of lighting portions **34** are provided corresponding to the plurality of insertion-receiving portions **28**, respectively. The plurality of lighting portions **34** are disposed at, for example, the front portion **108F**. Each lighting portion **34** includes a member that transmits light. For example, each lighting portion **34** includes a transparent member or a translucent member.

[0037] FIG. **3** is a diagram for explaining the internal configuration of the connector unit **112** according to the reference example.

[0038] The connector unit **112** further includes a plurality of connection terminals **30** and a plurality of light guide members **36**.

[0039] The plurality of connection terminals **30** are arranged corresponding to the plurality of insertion-receiving portions **28**, respectively. Each connection terminal **30** has conductivity. The plurality of connection terminals **30** are disposed on the inner side of the connector unit **112**. However, each connection terminal **30** can be brought into contact with the external terminal **104a** via the corresponding insertion-receiving portion **28**. By coming into contact with the external terminal **104a** inserted into the corresponding insertion-receiving portion **28**, each connection terminal **30** is electrically connected to the external terminal **104a**.

[0040] The plurality of light guide members **36** are arranged corresponding to the plurality of lighting portions **34**. Each light guide member **36** is, for example, a light pipe that is long in the front-rear direction. The plurality of light guide members **36** are disposed on the inner side of the connector unit **112**. However, at least a part of each light guide member **36** may extend to the inside of the communication processing unit **114**.

[0041] Each light guide member **36** transmits light emitted by the corresponding light emitting element **24** to the corresponding lighting portion **34**. Consequently, each lighting portion **34** can be lit by using light transmitted from the corresponding light guide member **36**.

[0042] Incidentally, in the I/O unit **108**, as the number of the light guide members **36** provided in the I/O unit **108** increases, it takes more time to assemble the light guide members **36** to the I/O unit **108**.

[0043] Based on the above, an I/O unit **10** according to an embodiment will be described below. However, explanations overlapping with those of the above-described reference example will be omitted as much as possible in the following description. Elements that have been described in the reference example are denoted by the same reference numerals as in the reference example unless otherwise specified.

[0044] FIG. **4A** is a perspective view showing a front portion **10F** and a left side portion **10L** of the I/O unit **10** according to the embodiment. FIG. **4B** is an exploded view of the I/O unit **10** of FIG. **4A**. Note that a right side portion **10R** of the I/O unit **10** may be substantially symmetrical with the left side portion **10L**.

[0045] The I/O unit **10** is preferably divisible into a base unit **12**, a communication processing unit **14**, and a connector unit **16**, which will be described below.

[0046] The base unit **12** is a part including a rear portion (a rear end portion) **10B** of the I/O unit **10**. The base unit **12** includes an end engagement portion **18**, and a plurality of contactors **20**.

[0047] The end engagement portion **18** is provided at the rear portion **10B**. The end engagement portion **18** is provided to engage with an installation member **110**. For example, the end engagement portion **18** includes a claw that engages with a part of the installation member **110**. The I/O unit **10** is installed on the installation member **110** by the end engagement portion **18** engaging with the installation member **110**.

[0048] The plurality of contactors **20** are disposed on each of a left side portion **12L** of the base unit **12** and a right side portion of the base unit **12**. The right side portion of the base unit **12** is not shown.

[0049] Note that the left side portion **12L** of the base unit **12** is included in the left side portion **10L** of the I/O unit **10**. Further, the right side portion of the base unit **12** is included in the right side portion **10R** of the I/O unit **10**.

[0050] A part of each contactor **20** disposed on the left side portion **12L** is exposed to the outside of the base unit **12** at the left side portion **12L**. Similarly, a part of each contactor **20** disposed on the right side portion of the base unit **12** is exposed to the outside of the base unit **12** at the right side portion.

[0051] The communication processing unit **14** is a part that can be attached to and detached from the base unit **12**. The communication processing unit **14** is disposed forward of the base unit **12**, and is mounted on the base unit **12** toward the rear side.

[0052] The communication processing unit **14** includes a housing **14a** and a control module **26**. The housing **14a** accommodates the control module **26**. The control module **26** includes a communication processing circuit **22** and a plurality of light emitting elements **24**.

[0053] The communication processing circuit **22** is electrically connected to the plurality of contactors **20** by the communication processing unit **14** being mounted on the base unit **12**. Each light emitting element **24** emits light under the control of the communication processing circuit **22**. Each light emitting element **24** includes, for example, an LED.

[0054] The connector unit (a terminal block) **16** is a part that can be attached to and detached from

the communication processing unit **14**. The connector unit **16** is disposed forward of the communication processing unit **14**, and is mounted on the communication processing unit **14** toward the rear side. The connector unit **16** includes the front portion **10F** of the I/O unit **10**.

[0055] The connector unit **16** includes a housing **16a**, a plurality of insertion-receiving portions **28**, a plurality of release operating elements **32**, and a plurality of lighting portions **34**. The housing **16a** may include an engagement mechanism for coupling with the housing **14a** of the communication processing unit **14**. The plurality of insertion-receiving portions **28**, the plurality of release operating elements **32**, and the plurality of lighting portions **34** are disposed at, for example, the front portion **10F**.

[0056] Each of the plurality of insertion-receiving portions **28** is a portion to which a device **104** (see FIG. **1**) is connected. Each of the insertion-receiving portions **28** includes, for example, an opening into which an external terminal **104a** of the device **104** is inserted. The opening is formed in the front surface of the housing **16a**.

[0057] Note that a plurality of connection terminals **30** respectively corresponding to the plurality of insertion-receiving portions **28** are disposed inside the housing **16a** (see also FIG. **5**). The device **104** is connected to the connection terminal **30** corresponding to the insertion-receiving portion **28** into which the external terminal of the device **104** is inserted.

[0058] The plurality of connection terminals **30** are electrically connected to the communication processing circuit **22** by the connector unit **16** being mounted on the communication processing unit **14**. Therefore, the device **104** is electrically connected to the communication processing circuit **22** via the connection terminal **30** corresponding to the insertion-receiving portion **28** into which the external terminal of the device **104** is inserted.

[0059] The number of the insertion-receiving portions **28** provided in the connector unit **16** is not particularly limited. The plurality of insertion-receiving portions **28** may be arranged in a plurality of rows at the front portion **10F**. For example, the plurality of insertion-receiving portions **28** (**28A**, **28B**) illustrated in FIGS. **4A** and **4B** are arranged in two rows at the front portion **10F**. In each row, the plurality of insertion-receiving portions **28** are arranged in the up-down direction. One row and the other row are different in position in the left-right direction.

[0060] In the following description, the insertion-receiving portions **28** included in the left row of the above-described two rows are also referred to as first insertion-receiving portions **28A**. Further, the insertion-receiving portions **28** included in the right row of the above-described two rows are also referred to as second insertion-receiving portions **28B**.

[0061] The plurality of release operating elements **32** are accommodated in the housing **16a**. However, a part of each release operating element **32** is exposed to the outside of the housing **16a** at the front portion **10F** so that the operator can operate the release operating element **32**. Further, the plurality of release operating elements **32** (**32A**, **32B**) include a plurality of first release operating elements **32A** and a plurality of second release operating elements **32B**.

[0062] The plurality of first release operating elements **32A** are disposed corresponding to the plurality of first insertion-receiving portions **28A**, respectively. In the illustration of FIGS. **4A** and **4B**, each first release operating element **32A** is disposed on the left side of the corresponding first insertion-receiving portion **28A**. Further, the first release operating elements **32A** are arranged in the up-down direction.

[0063] The plurality of second release operating elements **32B** are disposed corresponding to the plurality of second insertion-receiving portions **28B**, respectively. In the illustration of FIGS. **4A** and **4B**, each second release operating element **32B** is disposed on the right side of the corresponding second insertion-receiving portion **28B**. Further, the plurality of second release operating elements **32B** are arranged in the up-down direction.

[0064] The plurality of lighting portions **34** (**34A**, **34B**) include a plurality of first lighting portions **34A** and a plurality of second lighting portions **34B**.

[0065] The plurality of first lighting portions **34A** are disposed corresponding to the plurality of

first insertion-receiving portions **28A**, respectively. In the illustration of FIGS. **4A** and **4B**, each first lighting portion **34A** is disposed on the left side of the corresponding first insertion-receiving portion **28A**. Further, the plurality of first lighting portions **34A** are arranged in the up-down direction.

[0066] The plurality of second lighting portions **34B** are disposed corresponding to the plurality of second insertion-receiving portions **28B**, respectively. In the illustration of FIGS. **4A** and **4B**, each second lighting portion **34B** is disposed on the right side of the corresponding second insertion-receiving portion **28B**. Further, the second lighting portions **34B** are arranged in the up-down direction.

[0067] FIG. **5** is a diagram for explaining the internal configuration of the connector unit **16** according to the embodiment.

[0068] The connector unit **16** further includes a plurality of light guide members **36** and a plurality of coupling members **38**.

[0069] The plurality of light guide members **36** and the plurality of coupling members **38** are accommodated in the housing **16a**. However, at least a portion of each of the plurality of light guide members **36** and at least a portion of each of the plurality of coupling members **38** may protrude rearward from the housing **16a**. In this case, the portions of the plurality of light guide members **36** and the portions of the plurality of coupling members **38** that protrude rearward from the housing **16a** may be accommodated in the housing **14a** of the communication processing unit **14** by the connector unit **16** being mounted on the communication processing unit **14**.

[0070] Each of the plurality of light guide members **36** is long in the front-rear direction. Further, each light guide member **36** includes a first end portion **42** and a second end portion **44**. The first end portion **42** is a tip portion of the light guide member **36** on the front side. The second end portion **44** is a tip portion of the light guide member **36** on the rear side.

[0071] The first end portions **42** of the plurality of light guide members **36** are connected to the different lighting portions **34**. Further, the second end portions **44** of the plurality of light guide members **36** are connected to the different light emitting elements **24** by the connector unit **16** being mounted on the communication processing unit **14**.

[0072] Each light guide member **36** guides light emitted by the corresponding light emitting element **24** to the corresponding lighting portion **34**. As a result, the plurality of lighting portions **34** are lit in response to the light emission of the plurality of light emitting elements **24**.

[0073] According to the embodiment, the plurality of light guide members **36** (**36A**, **36B**) include the light guide member **36** connected to the first lighting portion **34A**, and the light guide member **36** connected to the second lighting portion **34B**. In view of this, the light guide member **36** connected to the first lighting portion **34A** is also referred to as a first light guide member **36A** in the following description. Further, the light guide member **36** connected to the second lighting portion **34B** is also referred to as a second light guide member **36B** in the following description.

[0074] Each of the plurality of coupling members **38** is a linear (columnar) member that is long in the up-down direction. Each of the coupling members **38** couples together the plurality of light guide members **36**. In this instance, the plurality of coupling members **38** (**38A**, **38B**) include a plurality of first coupling members **38A**, and a plurality of second coupling members **38B**.

[0075] Each of the plurality of first coupling members **38A** couples together the plurality of first light guide members **36A**. The plurality of first light guide members **36A** and the plurality of first coupling members **38A** may be integrally formed.

[0076] Each of the plurality of second coupling members **38B** couples together the plurality of second light guide members **36B**. The plurality of second light guide members **36B** and the plurality of second coupling members **38B** may be integrally formed.

[0077] The number of times of insertion operation required for the operator to accommodate the plurality of coupled light guide members **36** in the housing **16a** is only one. That is, according to the present embodiment, the operator does not need to repeat the operation of accommodating the

plurality of light guide members **36** in the housing **16a** the same number of times as the number of the light guide members **36**. Therefore, the operator can easily and quickly perform the assembly operation of the I/O unit **10**.

[0078] In this manner, according to the present embodiment, an increase in the operating time for assembling the plurality of light guide members **36** to the I/O unit **10** is suppressed.

[0079] Each coupling member **38** is preferably made of the same material as the light guide member **36**. This suppresses an increase in the number of types of materials included in the connector unit **16**.

[0080] Note that, in a case where the coupling member **38** is made of the same material as the light guide member **36**, there is a possibility that light leaks from the light guide member **36** to the coupling member **38**. The light leaking from the light guide member **36** to the coupling member **38** may reduce the amount of light of the lighting portion **34** corresponding to that light guide member **36**.

[0081] In view of this, it is preferable that the connection locations of the plurality of coupling members **38** with the light guide members **36** and the shapes of the light guide members **36** are designed in consideration of the characteristics of the material related to the reflection angle of light. This can minimize the possibility that light leaks from the light guide member **36** to the coupling member **38**.

[0082] Moreover, according to the present embodiment, the I/O unit **10** can be divided into a plurality of units. Specifically, the I/O unit **10** can be divided into the connector unit **16** having a function of connecting to the device **104**, the communication processing unit **14** having a function of performing communication processing, and the base unit **12** having a function of connecting to another I/O unit **10**. As a result, the operator can perform maintenance on each of the connector unit **16**, the communication processing unit **14**, and the base unit **12**, for example.

Modifications

[0083] Modifications of the above-described embodiment will be described below. However, explanations overlapping with those of the above-described embodiment will be omitted as much as possible in the following description. Elements that have been described in the above-described embodiment are denoted by the same reference numerals as in the above-described embodiment unless otherwise specified.

(Modification 1)

[0084] FIG. **6A** is a diagram showing a connector unit **161** (**16**) according to a first modification. FIG. **6B** is a diagram showing another connector unit **162** (**16**) according to the first modification.

[0085] All the insertion-receiving portions **28** may be arranged in a line (FIGS. **6A** and **6B**).

Similarly, all the lighting portions **34** may be arranged in a line.

[0086] The plurality of lighting portions **34** may be disposed on the right side or the left side of the plurality of insertion-receiving portions **28** arranged in a line. The lighting portions **34** disposed on the right side of the plurality of insertion-receiving portions **28** and the lighting portions **34** disposed on the left side of the plurality of insertion-receiving portions **28** may be mixed.

(Modification 2)

[0087] FIG. **7** is a diagram showing a connector unit **163** (**16**) according to a second modification.

[0088] The connector unit **163** includes the components of the connector unit **16** (see the embodiment). The connector unit **163** further includes an attachment member **40**.

[0089] The attachment member **40** has a size that can be accommodated inside the housing **16a**. For example, a width $W_{sub.40}$ of the attachment member **40** is smaller than a width $W_{sub.16}$ of the inner side of the housing **16a** in the left-right direction ($W_{sub.40} < W_{sub.16}$).

[0090] Further, a plurality of attachment portions **46** are formed in the attachment member **40**. Each attachment portion **46** is formed for attaching the light guide member **36** to the attachment member **40**. Each attachment portion **46** includes, for example, a groove portion capable of holding the light guide member **36**.

[0091] The plurality of attachment portions **46** are formed at a left end portion **40L** and a right end portion **40R** of the attachment member **40**. The plurality of attachment portions **46** formed at the left end portion **40L** hold the plurality of first light guide members **36A**. The plurality of attachment portions **46** formed at the right end portion **40R** hold the plurality of second light guide members **36B**.

[0092] The plurality of light guide members **36** attached to the attachment member **40** protrude forward from the attachment member **40**.

[0093] The operator accommodates, in the housing **16a**, the attachment member **40** to which the plurality of light guide members **36** are attached. Consequently, the operator can collectively accommodate all the light guide members **36** in the housing **16a** by one operation of accommodating the attachment member **40** in the housing **16a**.

[0094] At least a portion of the attachment member **40** may protrude rearward from the housing **16a**. In this case, the portion of the attachment member **40** that protrudes rearward from the housing **16a** may be accommodated in the housing **14a** of the communication processing unit **14** by the connector unit **16** being mounted on the communication processing unit **14**.

[0095] When the attachment member **40** is employed, it is more preferable that each of the plurality of light guide members **36** includes a bent portion **36c**. The bent portion **36c** of each light guide member **36** is located forward of the attachment member **40** in a state where the light guide member **36** is attached to the attachment member **40**.

[0096] Each light guide member **36** is bent in a substantially crank shape at the bent portion **36c**. The first end portion **42** of each light guide member **36** faces forward in front of the bent portion **36c**.

[0097] The way of bending at the bent portion **36c** is different between the first light guide member **36A** and the second light guide member **36B**. Specifically, each first light guide member **36A** is bent toward the second light guide member **36B** (rightward) at the bent portion **36c**. On the other hand, each second light guide member **36B** is bent toward the first light guide member **36A** (leftward) at the bent portion **36c**.

[0098] Accordingly, an interval $W_{sub.35}$ between the first end portion **42** of the first light guide member **36A** and the first end portion **42** of the second light guide member **36B** in the left-right direction becomes smaller than the width $W_{sub.40}$ of the attachment member **40** ($W_{sub.36} < W_{sub.40}$). Consequently, the interval $W_{sub.36}$ between the first end portion **42** of the first light guide member **36A** and the first end portion **42** of the second light guide member **36B** in the left-right direction can be made sufficiently smaller than the width $W_{sub.16}$ of the inner side of the housing **16a** ($W_{sub.36} < W_{sub.16}$). As a result, the assembly operation of the connector unit **16** is facilitated.

[0099] Specifically, the operator attaches the plurality of light guide members **36** to the above-described attachment member **40** in the assembly operation of the connector unit **16**. Next, the operator accommodates the attachment member **40** in the housing **16a** toward the front side from the rear of the housing **16a**.

[0100] Here, since the interval $W_{sub.36}$ is sufficiently smaller than the width $W_{sub.16}$, the operator can easily insert the first end portions **42** of the first light guide members **36A** and the first end portions **42** of the second light guide members **36B** into the inside of the housing **16a**.
(Modification 3)

[0101] The number of the first coupling members **38A** for coupling the plurality of first light guide members **36A** may be one. Similarly, the number of the second coupling members **38B** for coupling the plurality of second light guide members **36B** may be one.

(Modification 4)

[0102] The first end portion **42** of each light guide member **36** and the lighting portion **34** corresponding to the light guide member **36** may be integrally formed.

(Combination of Plurality of Modifications)

[0103] The modifications described above may be appropriately combined within a range in which no contradiction occurs.

Inventions Obtained from Embodiment

[0104] Hereinafter, the inventions that can be grasped from the above-described embodiment and modifications will be described.

First Aspect of Invention

[0105] A first aspect of the present invention is the terminal block **(16)** including: the plurality of insertion-receiving portions **(28)** into which the plurality of external terminals **(104a)** are inserted, respectively, and which are provided respectively with the connection terminals **(30)** each electrically connected to a corresponding one of the inserted external terminals; the plurality of lighting portions **(34)** provided corresponding to the plurality of insertion-receiving portions, respectively; the plurality of light guide members **(36)** provided corresponding to the plurality of lighting portions, respectively, and each configured to light a corresponding one of the plurality of lighting portions by guiding, to the corresponding one of the plurality of lighting portions, light emitted by a corresponding one of the plurality of light emitting elements **(24)**; and at least one coupling member **(38)** coupling the plurality of light guide members.

[0106] This suppresses an increase in the operating time for assembling the light guide members to the terminal block.

[0107] The plurality of insertion-receiving portions may be provided so as to be arranged along a first direction, and the plurality of light guide members may be provided so as to be arranged along the first direction at positions that are different from positions of the plurality of insertion-receiving portions in a second direction orthogonal to the first direction.

[0108] The plurality of insertion-receiving portions may include the plurality of first insertion-receiving portions **(28A)** arranged along the first direction, and the plurality of second insertion-receiving portions **(28B)** arranged along the first direction at positions that are different from positions of the plurality of first insertion-receiving portions in the second direction.

[0109] The plurality of light guide members may include the plurality of first light guide members **(36A)** provided on one side of the plurality of insertion-receiving portions in the second direction, and the plurality of second light guide members **(36B)** provided on the other side of the plurality of insertion-receiving portions in the second direction, and the coupling member may include at least one first coupling member **(38A)** coupling the plurality of first light guide members, and at least one second coupling member **(38B)** coupling the plurality of second light guide members.

[0110] The first aspect of the present invention may further include the attachment member **(40)**, the plurality of first light guide members being attached to the attachment member on one side in the second direction and the plurality of second light guide members being attached to the attachment member on the other side in the second direction, and the housing **(16a)** configured to accommodate the plurality of first light guide members and the plurality of second light guide members together with the attachment member, the plurality of first light guide members and the plurality of second light guide members may protrude from the attachment member in the direction toward which the first light guide members and the second light guide members are accommodated in the housing, each of the plurality of first light guide members may be bent toward the other side in the second direction, and each of the plurality of second light guide members may be bent toward the one side in the second direction. This improves the ease of assembly.

[0111] The at least one coupling member and the plurality of light guide members may contain the same material. This enables the plurality of coupling members and the plurality of light guide members to be integrally manufactured in the same manufacturing process.

Second Aspect of Invention

[0112] A second aspect of the present invention is the I/O unit **(10)** including the terminal block according to the first aspect, and the I/O unit includes the plurality of light emitting elements, and the communication processing circuit **(22)** configured to control input/output of the signals to/from

the plurality of external terminals connected to the terminal block, and control light emission of the plurality of light emitting elements.

[0113] This suppresses an increase in the operating time for assembling the light guide members to the I/O unit.

REFERENCE SIGNS LIST

[0114] **10**: I/O unit [0115] **16, 161 to 163**: connector unit (terminal block) [0116] **16a**: housing [0117] **22**: communication processing circuit [0118] **24**: light emitting element [0119] **26**: control module [0120] **28**: insertion-receiving portion [0121] **30**: connection terminal [0122] **34**: lighting portion [0123] **36**: light guide member [0124] **38**: coupling member [0125] **40**: attachment member [0126] **104a**: external terminal

Claims

1. A terminal block comprising: a plurality of insertion-receiving portions into which a plurality of external terminals are inserted, respectively, and which are provided respectively with connection terminals each electrically connected to a corresponding one of the inserted external terminals; a plurality of lighting portions provided corresponding to the plurality of insertion-receiving portions, respectively; a plurality of light guide members provided corresponding to the plurality of lighting portions, respectively, and each configured to light a corresponding one of the plurality of lighting portions by guiding, to the corresponding one of the plurality of lighting portions, light emitted by a corresponding one of a plurality of light emitting elements; and at least one coupling member coupling the plurality of light guide members.
2. The terminal block according to claim 1, wherein the plurality of insertion-receiving portions are provided so as to be arranged along a first direction, and the plurality of light guide members are provided so as to be arranged along the first direction at positions that are different from positions of the plurality of insertion-receiving portions in a second direction orthogonal to the first direction.
3. The terminal block according to claim 2, wherein the plurality of insertion-receiving portions include: a plurality of first insertion-receiving portions arranged along the first direction; and a plurality of second insertion-receiving portions arranged along the first direction at positions that are different from positions of the plurality of first insertion-receiving portions in the second direction.
4. The terminal block according to claim 2, wherein the plurality of light guide members include: a plurality of first light guide members provided on one side of the plurality of insertion-receiving portions in the second direction; and a plurality of second light guide members provided on another side of the plurality of insertion-receiving portions in the second direction, and the coupling member includes: at least one first coupling member coupling the plurality of first light guide members; and at least one second coupling member coupling the plurality of second light guide members.
5. The terminal block according to claim 4, further comprising: an attachment member, the plurality of first light guide members being attached to the attachment member on one side in the second direction and the plurality of second light guide members being attached to the attachment member on another side in the second direction; and a housing configured to accommodate the plurality of first light guide members and the plurality of second light guide members together with the attachment member, wherein the plurality of first light guide members and the plurality of second light guide members protrude from the attachment member in a direction toward which the first light guide members and the second light guide members are accommodated in the housing, and each of the plurality of first light guide members is bent toward the other side in the second direction, and each of the plurality of second light guide members is bent toward the one side in the second direction.
6. The terminal block according to claim 1, wherein the at least one coupling member and the

plurality of light guide members contain a same material.

7. An I/O unit comprising: the terminal block according to claim 1; the plurality of light emitting elements; and a communication processing circuit configured to control input/output of signals to/from the plurality of external terminals connected to the terminal block, and control light emission of the plurality of light emitting elements.
